

Groundwater

Problem 1: Solutions

a) Direction of groundwater flow

Since groundwater flows from high hydraulic head to low hydraulic head, and $h_1 = 145$ m $>$ $h_2 = 125$ m, flow is **downhill from point 1 to point 2**.

Explanation: The fundamental principle of groundwater flow is that water moves from areas of high hydraulic head to areas of low hydraulic head, regardless of topography. In this case, point 1 has a higher hydraulic head than point 2, so flow occurs from 1 to 2.

b) Hydraulic gradient calculation

The hydraulic gradient is calculated as:

$$i = \frac{h_1 - h_2}{\Delta l} = \frac{145 - 125}{5000} = \frac{20}{5000} = 0.004 \text{ m/m (dimensionless)}$$

Explanation: The hydraulic gradient represents the change in hydraulic head per unit distance along the flow path. It is dimensionless because it's a ratio of length units (meters of head change per meters of distance). This gradient drives groundwater flow according to Darcy's law.

c) Groundwater flow exit point

Flow leaves through the **downhill face of the aquifer** (at point 2, shown in blue in the figure). This is the discharge zone where groundwater emerges from the aquifer system.

Explanation: The discharge zone occurs where the aquifer intersects the land surface at topographically low areas. At point 2, the confined aquifer becomes exposed, allowing groundwater to emerge as springs or seepage zones. This is a common occurrence in hilly terrain where aquifers outcrop at valley bottoms.

Key Learning Points:

- Groundwater always flows from high to low hydraulic head
- Hydraulic gradient = change in head / distance

- The gradient is dimensionless (m/m)
- Discharge occurs where the aquifer intersects topographic lows

Problem 2: Solutions

a) Calculate linear groundwater velocity for granular porous medium

Given:

- Hydraulic gradient $i = 0.004$ (from Problem 1)
- Hydraulic conductivity $K = 1 \times 10^{-4}$ m/sec
- Total porosity $n_T = 0.3$
- For granular medium: effective porosity $n_e \approx n_T = 0.3$

Solution: Using Darcy's equation for linear velocity: $v = -\frac{K}{n_e} \frac{(h_1 - h_2)}{\Delta l} = \frac{K \cdot i}{n_e}$

$$v = \frac{1 \times 10^{-4} \times 0.004}{0.3} = \frac{4 \times 10^{-7}}{0.3} = 1.33 \times 10^{-6} \text{ m/sec}$$

Converting to m/day: $v = 1.33 \times 10^{-6} \times 86,400 = 0.115 \text{ m/day}$

b) Calculate linear groundwater velocity for fractured shale

Given:

- Same hydraulic gradient $i = 0.004$ and $K = 1 \times 10^{-4}$ m/sec
- Total porosity $n_T = 0.1$
- Fracture porosity $n_f = 3 \times 10^{-3}$
- For fractured medium: effective porosity $n_e = n_f = 3 \times 10^{-3}$

Solution: $v = \frac{K \cdot i}{n_e} = \frac{1 \times 10^{-4} \times 0.004}{3 \times 10^{-3}} = \frac{4 \times 10^{-7}}{3 \times 10^{-3}} = 1.33 \times 10^{-4} \text{ m/sec}$

Converting to m/day: $v = 1.33 \times 10^{-4} \times 86,400 = 11.5 \text{ m/day}$

Key Insight: If this groundwater was contaminated, the linear groundwater velocity through the fracture network would be **100 times faster** than if the medium was porous. This demonstrates why fractured rock aquifers pose greater risks for contaminant transport.

Additional Learning Points:

- Linear velocity depends on effective porosity, not total porosity
- In fractured media, flow occurs primarily through fractures
- Fracture porosity is typically much smaller than total porosity
- Higher velocities in fractured systems increase contaminant transport rates

Problem 3: Solutions**a) Total hydraulic head at the measurement point**

The total hydraulic head is measured at the **bottom of the piezometer** (the measurement point), which corresponds to the **water level in the piezometer**.

Calculation:

- Ground surface elevation = 1000 m
- Depth to water = 25 m
- **Total hydraulic head $h = 1000 - 25 = 975$ m above sea level**

Explanation: The total hydraulic head equals the elevation of the water level in the piezometer. This represents the total energy per unit weight of water at the measurement point.

b) Pressure head at the measurement point

The pressure head represents the height of water column above the measurement point.

Calculation: Using the relationship: $h = z + \frac{P}{\rho_w g}$

Where:

- z = elevation head at measurement point = $1000 - 50 = 950$ m
- h = total hydraulic head = 975 m

Pressure head $\frac{P}{\rho_w g} = h - z = 975 - 950 = 25$ m

Explanation: The pressure head is the difference between the total hydraulic head and the elevation head at the measurement point. It represents the pressure expressed as an equivalent height of water column.

c) **Pressure at the measurement point**

Using pressure head $P = \rho_w g \times \text{pressure head}$ $P = 1000 \text{ kg/m}^3 \times 9.81 \text{ m/s}^2 \times 25 \text{ m}$ $P = 245,250 \text{ Pa} = 0.245 \text{ MPa}$

Explanation: The pressure at the measurement point results from the weight of the 25-meter water column above it. Both calculation methods give essentially the same result (small difference due to rounding in the conversion factor).

Key Learning Points:

- **Total hydraulic head** = elevation of water level in piezometer
- **Pressure head** = height of water column above measurement point
- **Elevation head** = height of measurement point above datum
- **Relationship:** $h = z + \frac{P}{\rho_w g}$
- Pressure increases with depth due to hydrostatic pressure
- 1 meter of water $\approx 9806 \text{ Pa} \approx 0.0098 \text{ MPa}$

Problem 4: Solutions

Step 1: Convert hydraulic conductivity to m/sec

$$K = 15.24 \text{ m/day} = 15.24 \text{ m/day} \times \frac{1 \text{ day}}{24 \times 60 \times 60 \text{ sec}} = 1.76 \times 10^{-4} \text{ m/sec}$$

Step 2: Find water properties at 20°C and 1 atm

For water at 20°C and 1 atm pressure:

- Dynamic viscosity $\mu = 1.023 \times 10^{-7} \text{ m} \cdot \text{sec}$ (This is actually $1.023 \times 10^{-3} \text{ Pa} \cdot \text{s}$ or $1.023 \times 10^{-3} \text{ kg}/(\text{m} \cdot \text{s})$)
- Water density $\rho_w = 1000 \text{ kg/m}^3$
- Gravitational acceleration $g = 9.81 \text{ m/s}^2$

Step 3: Calculate intrinsic permeability using the relationship

The relationship between hydraulic conductivity and intrinsic permeability is: $K = \frac{k \rho_w g}{\mu}$

Solving for intrinsic permeability k : $k = \frac{K \mu}{\rho_w g}$

Step 4: Substitute values

$$k = \frac{(1.76 \times 10^{-4} \text{ m/sec}) \times (1.023 \times 10^{-3} \text{ Pa.s})}{(1000 \text{ kg/m}^3) \times (9.81 \text{ m/s}^2)}$$

$$k = \frac{1.8 \times 10^{-7} \text{ m.Pa.s/sec}}{9810 \text{ kg/(m.s}^2\text{)}}$$

Since $\text{Pa} = \text{kg}/(\text{m.s}^2)$, we have: $k = \frac{1.8 \times 10^{-7}}{9810} \text{ m}^2 = 1.8 \times 10^{-11} \text{ m}^2$

Alternative calculation using the given approach: $k = (1.023 \times 10^{-7} \text{ m.sec}) \times (1.76 \times 10^{-4} \text{ m/sec}) = 1.8 \times 10^{-11} \text{ m}^2$

Key Learning Points:

- **Intrinsic permeability** is a property of the porous medium only (independent of fluid properties)
- **Hydraulic conductivity** depends on both the medium and fluid properties
- **Relationship:** $K = \frac{k \rho_w g}{\mu}$ where k is intrinsic permeability
- Units: intrinsic permeability is in m^2 , hydraulic conductivity is in m/s
- Water viscosity varies with temperature (lower viscosity at higher temperatures)
- Intrinsic permeability is useful for comparing different porous media
- The same medium will have different hydraulic conductivities for different fluids

Problem 5: Solutions

a) Equivalent horizontal hydraulic conductivity K_h

For **horizontal flow** (parallel to layers), the equivalent conductivity is the **thickness-weighted arithmetic mean**:

$$K_h = \frac{\sum (b_i \times K_i)}{\sum b_i}$$

Where b_i is the thickness of each layer and K_i is the hydraulic conductivity of each layer.

Calculation: $K_h = \frac{(225 \text{ m})(10^{-5} \text{ m/sec}) + (75 \text{ m})(1.92 \times 10^{-12} \text{ m/sec})}{300 \text{ m}}$

$$K_h = \frac{2.25 \times 10^{-3} + 1.44 \times 10^{-10}}{300} = \frac{2.25 \times 10^{-3}}{300} = 7.5 \times 10^{-6} \text{ m/sec}$$

Note: The shale contribution is negligible compared to sandstone for horizontal flow.

b) **Equivalent vertical hydraulic conductivity K_v**

For **vertical flow** (perpendicular to layers), the equivalent conductivity is the **thickness-weighted harmonic mean**:

$$K_v = \frac{\sum b_i}{\sum \frac{b_i}{K_i}}$$

Calculation: $K_v = \frac{300 \text{ m}}{\frac{225 \text{ m}}{10^{-5} \text{ m/sec}} + \frac{75 \text{ m}}{1.92 \times 10^{-12} \text{ m/sec}}}$

$$K_v = \frac{300}{2.25 \times 10^7 + 3.91 \times 10^{13}} = \frac{300}{3.91 \times 10^{13}} = 7.7 \times 10^{-12} \text{ m/sec}$$

Note: The shale dominates the vertical flow behavior due to its very low permeability.

c) **Flow direction in the shale layer**

When flow crosses a boundary between materials with different hydraulic conductivities, **flow refraction** occurs according to Snell's law for groundwater:

$$\frac{\tan \theta_1}{K_1} = \frac{\tan \theta_2}{K_2}$$

Where θ is measured from the normal to the boundary (vertical direction in this case).

Given:

- $\theta_1 = 45^\circ$ (angle in sandstone from vertical)
- $K_1 = K_{ss} = 10^{-5} \text{ m/sec}$ (sandstone)
- $K_2 = K_{sh} = 1.92 \times 10^{-12} \text{ m/sec}$ (shale)

Calculation: $\frac{\tan 45^\circ}{10^{-5}} = \frac{\tan \theta_2}{1.92 \times 10^{-12}}$

$$\frac{1}{10^{-5}} = \frac{\tan \theta_2}{1.92 \times 10^{-12}}$$

$$\tan \theta_2 = \frac{1.92 \times 10^{-12}}{10^{-5}} = 1.92 \times 10^{-7}$$

$$\theta_2 = \arctan(1.92 \times 10^{-7}) \approx 1.1 \times 10^{-5} \text{ degrees} \approx 0^\circ$$

Result: Flow in the shale is essentially **vertical** (almost parallel to the normal), demonstrating that low-permeability layers tend to channel flow in the direction normal to the layers.

Key Learning Points:

- **Horizontal equivalent conductivity:** arithmetic mean (high-K layers dominate)
- **Vertical equivalent conductivity:** harmonic mean (low-K layers dominate)
- **Flow refraction:** flow bends toward the normal when entering low-permeability material
- **Snell's law for groundwater:** $\frac{\tan \theta_1}{K_1} = \frac{\tan \theta_2}{K_2}$
- **Layered systems:** create strong anisotropy ($K_h \gg K_v$ when low-K layers present)
- **Practical implication:** contaminants preferentially move horizontally in layered aquifers